Physical accuracy beyond plausible dynamic rendering

Research supplement | Simulation study | 2 November 2026

Research question

A model can reconstruct observed motion and render an attractive scene while predicting the wrong response to a new load. We test these two abilities separately: future physical response under larger drops, and image quality from observed and new camera positions. Better physical prediction with comparable or slightly worse rendering is the target outcome, not a premise of the evaluation.

On six paired forecasts across three fresh worlds, mean motion error is 1.15 mm versus 13.77 mm for native PhysTwin; deformation error is 1.09 versus 2.94 mm. Held-view foreground PSNR is 18.73 versus 12.84 dB. Our forecasts complete eight of eight cases; native completes six. These conditional results use the same initial appearance and finite-resolution synthetic references.

Representation and prediction

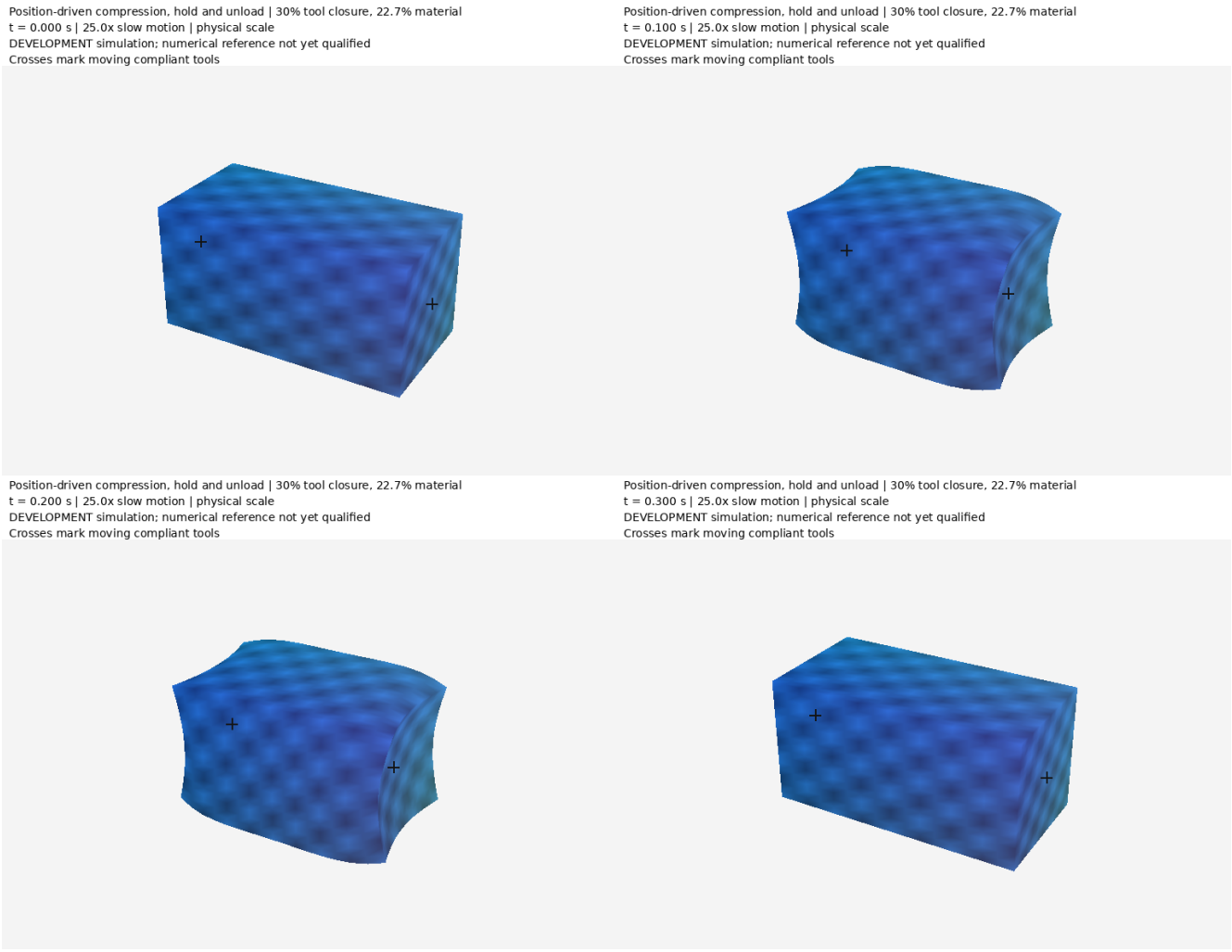
The representation separates material response, interactions and appearance. A shared neural bulk-energy model maps deformation and material descriptors to strain energy. Spatial material points move through a consistent-mass finite-element dynamic solver. Contact geometry supplies normal contact forces through a finite compliant law; prescribed tools supply position boundary conditions. A Gaussian appearance model is attached to material coordinates and transported with the deformation gradient.

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U(x,z) = integral W(F(x), z) dV + U_contact + U_tools
M x_ddot = -dU/dx + M g
mu_t = x_t(q) + F_t(q) offset
Sigma_t = F_t(q) Sigma_0 F_t(q)^T
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The common experiment observes a short 10 mm drop. Its inputs are known initial geometry, total mass, gravity, and visible 3D surface positions; neither method receives the material family, its parameters, the finite contact stiffness, or future motion. Our bounded inverse search selects family, scale, shape and contact compliance using only those observations. The fitted model is then reset and predicts a higher drop without refitting.

This study uses noiseless oracle 3D tracks with raster visibility, not end-to-end reconstruction from raw video. The global constitutive network was pretrained in earlier material-law studies; that prior is distinct from per-scene calibration. This extension does not establish calibrated posterior uncertainty or an active-probing advantage.

Visible deformation and actual dynamics



Recorded compression, hold and unload states. The prescribed tools close by 30% of the rest length; the material shortens by 22.7% because the attachments are compliant. The block bulges and recovers. Support crosses are tool annotations, not additional collision geometry. The video caption predates the final numerical audit; the smooth-cycle qualification below is now complete.

Example	Observed behavior	Evidence scope
Position-controlled cycle	22.7% material shortening; recovery	Six mesh/time/quadrature states pass
Finite 80 g striker at 1 m/s	Local bending and rebound; about 13.6% central compression	Coupled dynamic demonstration; not spatially qualified
40 mm viscous drop, 0.75 s	Impact, rebound and settling	Separate dissipative scenario; refinement reported separately

Videos show saved integrator states at physical scale. Slow motion is labeled; displacement is not magnified. Striker and compression demonstrations are not presented as learned-prediction benchmark wins.

Numerical evidence and framework controls

Refinement errors below use coordinate RMS. Forecast tables use vector RMS, which is sqrt(3) times coordinate RMS for the same weights.

Reference check	Measured difference	Conclusion
Smooth compression n3 to n4	53.2 um world/shape coordinate RMS	Pass for this controlled cycle
20 mm conservative drop n6 to n8	143.45 um world; 141.66 um shape	Full declared refinement conjunction passes
Abrupt 20% release n6 to n8	211.08 um shape; 150 um gate	Unqualified; failure retained
Stronger abrupt release/drop exploration	Four guard failures; finite striker completes	No relaxed guards or replacement cases

Small energy drift alone does not qualify a simulation. The drop and controlled-cycle checks compare time step, spatial resolution, quadrature and orientation preservation; contact-event checks additionally apply to the drop. The controlled cycle has no floor impact. The qualified 20 mm stable Neo-Hookean drop does not automatically qualify the new MR/Yeoh materials or higher impacts. The fresh study therefore reports finite-resolution teacher sensitivity explicitly.

Native Newton 1.7 / Warp 1.18 controls and the actual PhysTwin author simulator were executed. Newton development runs expose unresolved mesh/time and numerical-dissipation effects and are not used as a superior truth source. The main inferred-physics comparison uses author PhysTwin, with unchanged force and integration functions, known SI masses and a preserved fitted spring graph.

Native calibration configuration	Coordinate RMSE (mm)
125 points, 20 CMA / 200 epochs	1.32349
729 points, 20 CMA / 200 epochs	1.32383
125 points, 100 CMA / 1000 epochs	1.31495

The final baseline is globally selected from calibration replay only: 125 points, at most 100 CMA generations and 1000 gradient epochs. The 729-point capacity/configuration control and original 20/200 run remain reported. Logged online training loss is not substituted for fixed-checkpoint replay error. No further recipe selection uses the fresh targets.

Both methods receive the same total mass, but native uniform particle masses and continuum consistent mass differ. The 125-point grid has 1.5 times the continuum rest rotational inertia; 729 points have 1.25 times. Any prediction advantage is a configuration-level result that can jointly reflect representation, mass/contact discretization, material prior and inference.

Fresh prediction and rendering protocol

Four new worlds use balanced Mooney-Rivlin and Yeoh material families, privately drawn stiffness/shape/contact settings, a fixed 12 x 6 x 6 cm block, 0.27 kg mass and gravity 9.8 m/s^2. Each supplies 21 calibration frames over 0.20 s. The two forecasts are 40 mm and 80 mm drops, each 0.30 s with 31 common frames. This is load-magnitude extrapolation; it is not a new interaction-type or geometry benchmark.

Primary references use n4 discretization. A prospectively selected subset receives n6 and half-step sensitivity checks. All original worlds, failed prefixes and unavailable predictions remain in the eight-forecast census. There are four independent objects, not eight. Primary paired means require all eight valid complete forecasts; otherwise coverage and explicitly conditional complete-case summaries are shown.

Physical prediction	Rendering quality
Rest-area weighted world-position vector RMS	Foreground-union PSNR and SSIM
Shape RMS after proper rigid alignment only	Silhouette IoU
Nonrigid deformation and surface-centroid error	Joint-visible depth RMSE and coverage
Sampled penetration, closed-surface volume and bounce events	Observed-camera future frames and new camera views

The appearance comparison starts both methods from the same initial learned Gaussian checkpoint, trained only on initial views. PhysTwin uses its native K16 binding; our method uses material-coordinate centers and F Sigma F^T covariance transport. The common renderer, opacity rule, cameras and backgrounds are fixed. Thus image differences reflect motion and transport jointly. A separate procedural rendering isolates geometry without claiming native appearance parity.

All method/query physics attempts must be committed before target arrays are opened. All predicted image slots must be committed before reference images are generated. A 2 dB foreground PSNR or 0.02 SSIM difference is a descriptive practical band, not a statistical equivalence test. Physical and image scores remain separate; no composite score can hide a physically wrong response.

Development: motion and rendering

Development method	World RMS (mm)	Shape RMS (mm)
Ours, coarse n2	1.5749	1.3104
Ours, fine n3	0.2746	0.2156
Native 125 points, 20/200	9.7421	2.3349
Native 729 points, 20/200	9.0272	2.5720
Native 125 points, 100/1000 (selected)	9.5181	2.3036

One development object, with a same-n3 finite-resolution teacher for the fine variant. This is not a population or independently mesh-qualified prediction result. All variants are retained; the primary native configuration was selected using calibration only.

Prediction / views	Foreground PSNR (dB)	Foreground SSIM	IoU
Ours / observed views	26.55	0.940	0.990
Ours / new views	19.38	0.765	0.962
Native PhysTwin / observed views	16.16	0.679	0.858
Native PhysTwin / new views	13.82	0.598	0.843

Both methods use the same initial Gaussian appearance and native renderer. These scores measure motion plus transport, not independently reconstructed appearance. Frames/views are correlated; no population confidence interval is inferred from one object.

A post-hoc mechanism diagnostic removes only the selected baseline's fitted drag and dashpot, without refitting. World error falls to 6.147 mm, but shape error rises to 2.761 mm. Damping therefore explains part of the motion discrepancy, not the entire gap. This diagnostic does not replace the selected baseline or change the fresh cohort. Our elastic-only prior matches the teacher's zero physical damping; the comparison is not an isolated test of material-law superiority.

Fresh cohort: physical prediction

World / drop	Ours world mm	Native world mm	Ours shape mm	Native shape mm
world-0 / 40mm	0.943	8.815	0.858	2.437
world-0 / 80mm	1.180	19.424	1.065	2.879
world-1 / 40mm	1.150	8.530	1.130	3.274
world-1 / 80mm	1.485	17.720	1.462	3.751
world-2 / 40mm	0.968	8.829	0.893	2.355
world-2 / 80mm	1.151	19.309	1.105	2.918
world-3 / 40mm	0.819	unavailable	0.799	unavailable
world-3 / 80mm	1.010	unavailable	0.962	unavailable

Fresh method	Completed	Paired coverage	All-eight paired world RMS(mm)
Ours	8/8	6/8	unavailable
Native PhysTwin	6/8	6/8	unavailable

Unavailable is a failed or missing prediction/reference, never zero error. Each physical metric uses the complete saved sequence at matching times; only the shape metric removes proper rigid motion. The eight rows are two loads on each of four worlds, not eight independent objects.

Ours: on the same 6/8 available paired forecasts only, mean world error is 1.146 mm and mean shape error is 1.085 mm. These are conditional summaries; they do not substitute for an all-eight result.

Native PhysTwin: on the same 6/8 available paired forecasts only, mean world error is 13.771 mm and mean shape error is 2.936 mm. These are conditional summaries; they do not substitute for an all-eight result.

Reference sensitivity: 20/20 declared simulations completed. Spatial checks passed 0/4; time-step checks passed 4/4. The subset checks the 80 mm drop at n4/n6 and both time steps, plus calibration spatial consistency. It does not provide a continuum solution.

Native world-3 fitting stopped during graph initialization, leaving both forecasts unavailable. Our four selected fits used the fixed calibration budget; only one of the sixteen coarse/fine family optimizations reported convergence. Completed forecasts and optimizer convergence are distinct outcomes.

Reference-resolution robustness

After the declared spatial checks failed, a secondary diagnostic evaluates the same frozen predictions against the already generated finer references. It uses both prospectively fixed subset worlds and the 80 mm load. This is post-hoc analysis, with no refitting, new simulations or replacement of the primary n4 scores.

World / reference	Ours world mm	Native world mm	Ours shape mm	Native shape mm
world-0 / n4 base	1.180	19.424	1.065	2.879
world-0 / n6 base	1.690	18.779	1.059	2.858
world-0 / n6 halfstep	1.697	18.771	1.059	2.857
world-1 / n4 base	1.485	17.720	1.462	3.751
world-1 / n6 base	1.660	17.005	1.444	3.658
world-1 / n6 halfstep	1.659	17.011	1.446	3.659

Absolute errors depend on reference resolution. These rows permit a direct check of whether the method ordering changes with finer space/time discretization. A stable ordering does not repair the failed spatial qualification or establish continuum accuracy. The experiment remains a comparison of matched-observation configurations on finite-resolution synthetic dynamics.

Fresh cohort: new-view image prediction

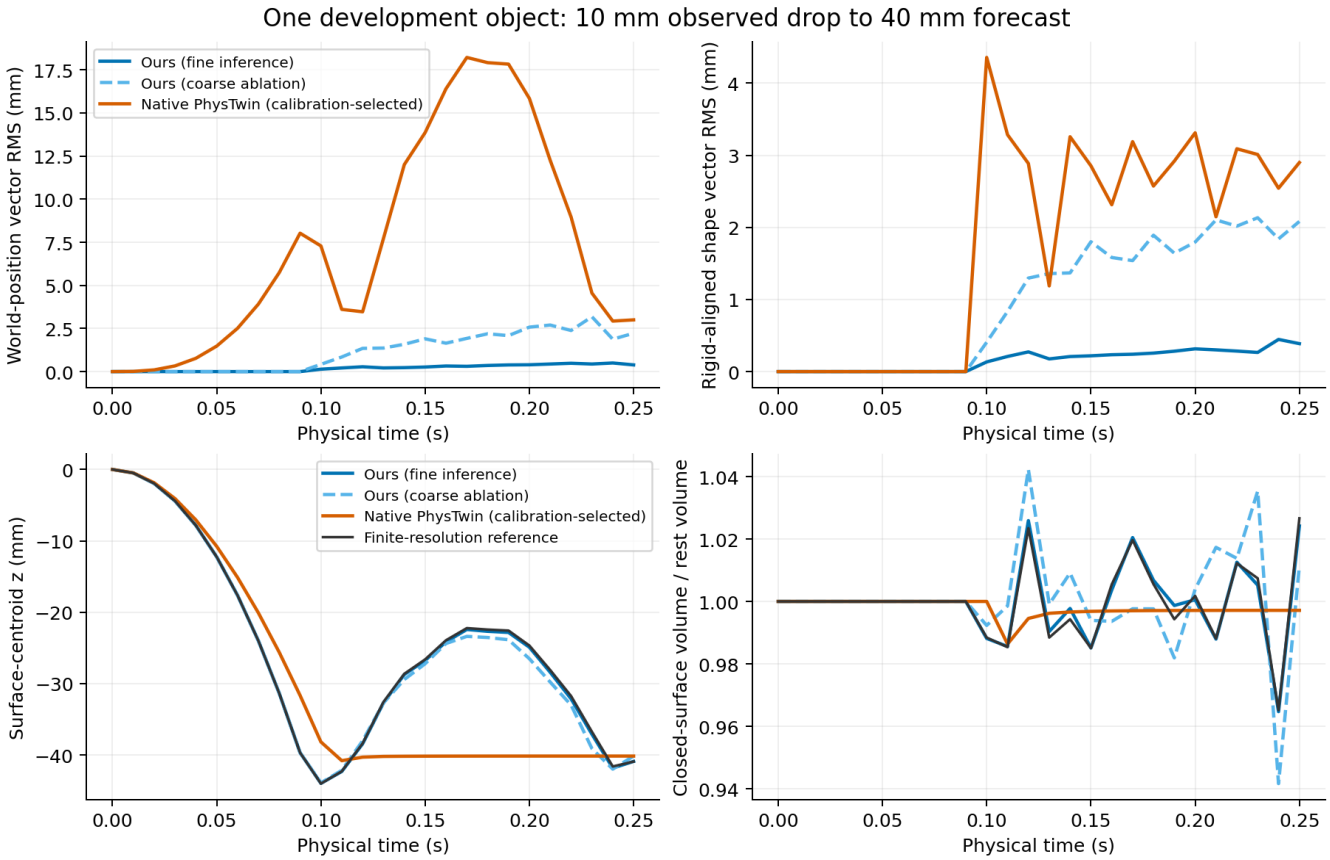
World / method	Scored / expected	Foreground PSNR	Foreground SSIM	IoU
world-0 / Native	124/124	12.885	0.580	0.794
world-0 / Ours	124/124	18.769	0.748	0.956
world-1 / Native	124/124	12.727	0.571	0.797
world-1 / Ours	124/124	18.650	0.745	0.955
world-2 / Native	124/124	12.913	0.580	0.794
world-2 / Ours	124/124	18.784	0.749	0.957
world-3 / Native	0/124	unavailable	unavailable	unavailable
world-3 / Ours	124/124	18.900	0.754	0.958

Each row averages both forecast loads, all 31 times and the two fixed held cameras within one world. Full-coverage values are shown; missing images remain unavailable. Higher PSNR/SSIM/IoU is better. Per-frame values, observed-camera scores, depth error and coverage are retained in the complete image report.

Both methods use the same initial Gaussian checkpoint and native rasterizer, with their respective motion/shape transport. This measures future-view prediction quality jointly with dynamics. It does not establish equality of independently learned appearance models. Correlated image frames are not treated as independent statistical samples.

Reproduction: reports/compositional-dynamics-v17 contains frozen protocols, source/input hashes, audit records and NATIVE-SETUP.txt. Raw simulation datasets and model checkpoints remain local. This supplement extends the preserved v16 manuscript rather than replacing its results.

Development response through impact and rebound



Top: unaligned world error and separately rigid-aligned shape error. Bottom: surface centroid (not mass center of mass) and the signed closed-surface volume ratio. The selected native model falls too slowly before impact and misses the rebound; the fine shared model tracks the reference closely. The coarse shared ablation retains a larger deformation error. All methods use the same recorded times; there is no temporal alignment.

The sampled-volume diagnostic does not certify interior orientation or energy conservation. Reference and method-specific energy/Jacobian checks are reported separately. The damping intervention is explanatory and post-hoc; no further model tuning follows the development outcome. The fresh four-world study keeps the prospectively frozen recipe.

Sources and reproducibility

[1] Hanxiao Jiang et al. PhysTwin: Physics-Informed Reconstruction and Simulation of Deformable Objects from Videos. ICCV 2025. Author implementation, pinned commit 81c718790a37e5e0102eb77af2c6edd34a9db25f.

[2] Newton developers. Newton 1.7 VBD solver documentation. The executed runtime is pinned separately in this project.

[3] Project protocols, raw-to-report hashes, independent audits and reproduction commands: reports/compositional-dynamics-v17. Earlier method and benchmark evidence: papers/compositional-neural-mechanics-v16. New raw datasets and trained checkpoints remain local.